

Cost-Aware Capacity Reservation for Cloud Workloads with Intermittent Random Demand Surges: An Integrated Forecasting–Optimization System

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Abstract

Enterprise cloud users face rapidly growing compute bills driven by workloads that combine a stable, seasonal *base* demand with *intermittent, random demand surges*. We study how such a user should provision capacity across the on-demand, reserved, and spot purchasing options offered by infrastructure-as-a-service providers, and we instantiate the resulting policy in a working software system. On the modeling side, we formulate the baseline reservation decision as a newsvendor problem and show that the cost-minimizing reserved capacity equals the δ -quantile of the demand distribution, where δ is the reserved-instance discount; the optimal reserve-versus-on-demand decision for a workload reduces to a simple utilization threshold $u^* = 1 - \delta$, and intermittent surges are best covered by *supplementary* reservations only when the surge time-fraction exceeds $1 - \delta_s$. On the systems side, we present *Smart Cloud Optimizer*, which ingests provider telemetry, forecasts demand with classical and machine-learning time-series models, and solves a mixed-integer program for instance right-sizing, complemented by rule-based and large-language-model recommendations. On a year-long, realistic workload, exponential smoothing forecasts demand at 12.1% MAPE; the newsvendor policy attains its theoretical optimum (verified against exhaustive search) and reduces cost by 26.7%–50.0% relative to pure on-demand provisioning depending on the discount tier; and the end-to-end system identifies \$590/month (54.9% of the addressable spend) in concrete savings. Our results connect the operations-management theory of capacity reservation to the engineering practice of cloud cost optimization, and show that a widely used industry heuristic—reserve long-running instances—is a special case of the optimal threshold policy.

Keywords: cloud computing; capacity reservation; newsvendor model; reserved instances; demand forecasting; mixed-integer programming; cost optimization.

1 Introduction

Public cloud spending has become one of the largest and fastest-growing line items in enterprise IT budgets, and a substantial fraction of it is wasted on idle or over-provisioned capacity. The discipline that has emerged to manage this spend—*FinOps*—emphasizes continuous visibility, right-sizing, and judicious use of provider pricing options (Storment and Fuller, 2023; Deochake, 2023). The central difficulty is that the economically optimal purchasing decision depends jointly on (i) the shape and uncertainty of the workload’s demand and (ii) the rich menu of pricing mechanisms that

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providers expose: pay-as-you-go *on-demand* instances, deeply discounted *reserved* instances (RIs) that require a one- or three-year commitment, and very cheap but interruptible *spot* instances (Armbrust et al., 2010; Kumar et al., 2018).

A recurring structural feature of real workloads sharpens the problem. Demand is rarely stationary: it typically consists of a predictable, seasonal *base* level punctuated by *intermittent*, *random surges* triggered by planned launches, marketing events, or exogenous traffic spikes (Chen et al., 2024). Reserving enough capacity to cover the surges leaves expensive capacity idle most of the time; reserving only the base forces the user onto premium on-demand pricing exactly when demand is highest. The user must therefore decide *how much* baseline capacity to reserve, and *how* to cover the residual surges.

In practice, this decision is usually made with crude heuristics. A common rule of thumb, embedded in many cost-optimization tools, is “reserve any instance that has been running continuously for more than 60 days.” Such rules are easy to implement but disconnected from the underlying economics: they ignore the discount rate, the demand distribution, and the intermittency of surges. At the same time, a rigorous operations-management literature on capacity reservation exists but is rarely connected to the engineering systems that actually issue cloud purchasing recommendations (Chen et al., 2024; Qu et al., 2024; Chen et al., 2023). This paper bridges that gap.

Contributions. We make four contributions.

1. **A capacity-reservation model for cloud workloads with intermittent surges.** We model a user who provisions a seasonal base plus random surges across reserved, on-demand, and spot options. We prove that the cost-minimizing baseline reservation follows a newsvendor (critical- fractile) solution: the optimal reserved level is the δ -quantile of demand, where δ is the RI discount (Proposition 1).
2. **A threshold characterization that rationalizes industry practice.** We show that the optimal reserve-versus-on-demand decision for a workload is a utilization threshold, $u^* = 1 - \delta$ (Proposition 2), and that supplementary reservation of surge capacity is optimal only when the surge time-fraction exceeds $1 - \delta_s$ (Proposition 3). The ubiquitous “reserve long-running instances” heuristic is exactly the special case $u \rightarrow 1$.
3. **An integrated, end-to-end system.** We present *Smart Cloud Optimizer*, which couples demand forecasting (five time-series models), a mixed-integer right-sizing program, rule-based service checks, and a large-language-model advisor, all over a unified telemetry store.
4. **A reproducible numerical study.** On a year-long, realistic workload exhibiting exactly the base-plus-surge structure our model assumes, we validate the theory (the closed-form reservation level matches exhaustive search to within \$0.03/day), quantify forecasting accuracy (best MAPE 12.1%), and report concrete end-to-end savings (\$590/month, 54.9% of the addressable spend).

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 sets up the model. Section 4 derives the optimal reservation and threshold policies. Section 5 describes the Smart Cloud Optimizer system. Section 6 reports the numerical study. Section 7 distills managerial insights, and Section 8 concludes. Proofs are collected in Appendix A.

2 Related Work

Our work draws on, and connects, several literatures.

Cloud cost optimization and pricing. The economics of elastic, pay-as-you-go computing were articulated early by [Armbrust et al. \(2010\)](#); the FinOps movement has since operationalized cost governance in practice ([Storment and Fuller, 2023](#); [Deochake, 2023](#)). Provider pricing mechanisms—on-demand, reserved, and spot—and their trade-offs have been surveyed extensively ([Kumar et al., 2018](#); [Mazrekaj et al., 2016](#); [Sharma et al., 2012](#)). [Wang et al. \(2013\)](#) design online algorithms that dynamically mix reserved and on-demand instances under time-varying demand; we instead characterize the static-commitment optimum in closed form and embed it in a forecasting-driven system.

Capacity provisioning under uncertainty. A rich body of work formulates cloud provisioning as optimization under demand and price uncertainty. [Chaisiri et al. \(2012\)](#) propose the canonical stochastic-programming model trading reserved against on-demand capacity; [Bülbül et al. \(2021\)](#) extend this to multi-stage stochastic programs, and [Mireslami et al. \(2021\)](#) and [Furman and Diamant \(2025\)](#) study dynamic and periodic time-varying demand. Relative to these, we obtain an interpretable newsvendor characterization tailored to the base-plus-surge structure and connect it directly to a deployed system.

Workload forecasting. Demand forecasting is the input to any provisioning policy. [Masdari and Khoshnevis \(2020\)](#) survey and classify cloud workload forecasting methods. Classical time-series approaches such as ARIMA remain strong baselines ([Calheiros et al., 2015](#)), while machine-learning methods—neural networks ([Kumar and Singh, 2018](#)), LSTMs ([Song et al., 2018](#)), and probabilistic deep models such as DeepAR ([Salinas et al., 2020](#))—target complex patterns. Our system implements a spectrum from naive baselines to Prophet and auto-tuned SARIMAX, and—critically—feeds the resulting predictive *distribution* into the newsvendor reservation policy.

Newsvendor and reservation contracts. The newsvendor model is the workhorse for single-period capacity decisions under uncertainty ([Qin et al., 2011](#)). Reservation contracts with later exercise have long been studied in operations management—e.g., [Erkoc and Wu \(2005\)](#) on high-tech capacity expansion in the same journal as our reference paper. Most directly, [Chen et al. \(2024\)](#) study capacity reservation for cloud workloads with intermittent random surges, deriving a newsvendor-type capacity and a two-threshold purchase/renew/cancel policy, and [Qu et al. \(2024\)](#) analyze long-run-average cloud cost with reserved and on-demand capacity. [Chen et al. \(2023\)](#) survey cloud value chains from an operations-management perspective. We adopt this modeling lens but specialize it to a discount-parameterized closed form and pair it with a working implementation.

Right-sizing and resource allocation. [Lin et al. \(2013\)](#) give a foundational online treatment of dynamically right-sizing data-center capacity; [Hadary et al. \(2020\)](#) describe a hyperscale VM-allocation system. Cost-optimal placement via integer programming appears in [Van den Bossche et al. \(2010\)](#). Our right-sizing module solves a mixed-integer program in this tradition, but as one component of a larger forecast-then-reserve pipeline.

3 Model

We consider a single cloud user (a firm) that must serve a compute workload over a planning horizon of T periods (days, in our implementation). All quantities are expressed in *capacity units*; without loss of generality we take the on-demand price as the numéraire, $p = 1$, so that demand and cost are measured on the same scale. Table 1 summarizes notation.

Table 1: Notation.

Symbol	Meaning
D_t	demand (required capacity) in period t , $t = 1, \dots, T$
F, f	cumulative distribution / density of demand D
p	on-demand unit price per period (numéraire, $p = 1$)
$\delta \in (0, 1)$	reserved-instance discount; reserved price $r = (1 - \delta)p$
δ_s	spot discount; spot price $s = (1 - \delta_s)p$, with $\delta_s > \delta$
$\beta \in [0, 1]$	spot interruption (reclaim) probability
$Q \geq 0$	reserved baseline capacity (decision variable)
d_b	deterministic base demand level
$\rho \in [0, 1]$	long-run fraction of time the workload is in a surge state
$u \in [0, 1]$	utilization: fraction of the term a unit of demand is active
$C(\cdot)$	total provisioning cost over the horizon

Demand structure. Motivated by real workloads (Section 6), we model demand as a stationary, seasonal *base* component plus *intermittent random surges*:

$$D_t = d_b + S_t M_t, \quad (1)$$

where $S_t \in \{0, 1\}$ indicates a surge in period t and $M_t \geq 0$ is the random surge magnitude. Surges are intermittent: $\Pr(S_t = 1)$ is small and surges occur in episodes separated by inter-surge periods. We let $\rho = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_t S_t$ denote the long-run surge time-fraction. The base d_b is predictable and seasonal (e.g., weekly), whereas the magnitude M_t is random.

Procurement options. In each period the user serves D_t from three sources, mirroring the AWS menu:

- **Reserved instances** (base contracts): a committed quantity Q , paid at the discounted rate $r = (1 - \delta)p$ *every period regardless of use*.
- **On-demand**: any quantity, paid at $p = 1$ only for what is used, always available.
- **Spot**: paid at $s = (1 - \delta_s)p < r$ when available, but reclaimed with probability β , in which case the load falls back to on-demand.

Reserved capacity is consumed first (it is sunk), then spot (when permitted by reliability requirements), then on-demand. The single-period cost of committing Q reserved units and covering the residual with on-demand is

$$c_t(Q) = \underbrace{(1 - \delta) Q}_{\text{reserved (always paid)}} + \underbrace{p (D_t - Q)^+}_{\text{on-demand top-up}}, \quad (x)^+ \equiv \max\{x, 0\}. \quad (2)$$

Assumption 1. Prices are ordered $0 < s < r < p$ with $r = (1 - \delta)p$, $s = (1 - \delta_s)p$, and $0 < \delta < \delta_s < 1$. The horizon is long enough that the empirical demand distribution $\hat{F}$ is a consistent estimate of F .

The user's problem is to choose the reservation level Q (and, in the extensions, supplementary and spot quantities) to minimize total cost $C(Q) = \sum_{t=1}^T c_t(Q)$, or equivalently the expected per-period cost $\mathbb{E}[c(Q)] = (1 - \delta)Q + p \mathbb{E}(D - Q)^+$.

4 Optimal Capacity Reservation

4.1 Baseline reservation: a newsvendor solution

We first solve for the cost-minimizing reserved level Q , taking the demand distribution F as given (it is supplied by the forecasting module of Section 5).

Proposition 1 (Optimal reserved capacity). *Under Assumption 1, the expected per-period cost $\mathbb{E}[c(Q)] = (1 - \delta)Q + p \mathbb{E}(D - Q)^+$ is convex in Q , and the unique cost-minimizing reservation level is the δ -quantile of demand,*

$$Q^* = F^{-1}(\delta) = \inf\{q : F(q) \geq \delta\}. \quad (3)$$

Equivalently, Q^* is the newsvendor solution with critical ratio $\text{CR} = \frac{C_u}{C_u + C_o} = \delta$, where the underage cost is $C_u = p - r = \delta p$ (the on-demand premium paid when reserved capacity is short) and the overage cost is $C_o = r = (1 - \delta)p$ (the cost of an idle reserved unit).

The intuition is transparent: a larger discount δ lowers the cost of holding reserved capacity, so the user reserves a higher quantile of demand. The result also delivers a clean comparative static.

Corollary 1 (Discount monotonicity). *Q^* is nondecreasing in the discount δ . The fraction of demand periods fully covered by reserved capacity, $F(Q^*)$, equals δ .*

Thus a firm should reserve *more* capacity, covering a higher demand quantile, precisely when reserved instances are more deeply discounted—a conclusion that aligns with [Chen et al. \(2024\)](#), who find that firms buy more supplementary capacity as the unit discount rate increases.

4.2 When to reserve: a utilization threshold

Proposition 1 prices the *quantity* of reservation. We now ask the dual, per-workload question that cost-optimization tools actually face: *should a given workload be on a reserved or an on-demand plan?* Consider a unit of demand that is active a fraction u of the reservation term (its *utilization*). Reserving it costs $r = (1 - \delta)p$ for the entire term; serving it on-demand costs p per active period, i.e., pu in expectation over the term.

Proposition 2 (Reservation break-even). *A unit of demand with utilization u should be reserved if and only if*

$$u \geq u^* = 1 - \delta. \quad (4)$$

At the real AWS discount tiers ($\delta = 0.37$ for one-year and $\delta = 0.60$ for three-year RIs), the break-even utilizations are $u^ = 0.63$ and $u^* = 0.40$, respectively.*

Proposition 2 rationalizes—and corrects—the industry heuristic. The popular rule “reserve an instance that has run continuously for ≥ 60 days” uses elapsed running time as a *proxy* for high utilization. The optimal rule replaces the proxy with the economically correct threshold $u^* = 1 - \delta$: an always-on workload ($u \rightarrow 1$) should always be reserved (since $1 > 1 - \delta$), but a workload active only, say, half the term should be reserved only under the deeper three-year discount ($0.5 \geq 0.40$) and *not* under the one-year discount ($0.5 < 0.63$). The 60-day rule is the $u \rightarrow 1$ corner of (4).

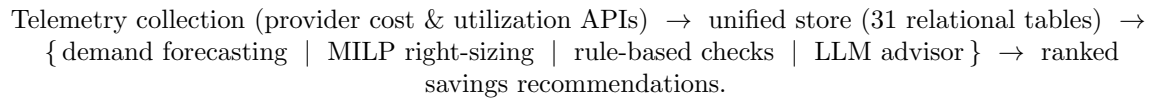


Figure 1: Smart Cloud Optimizer pipeline. Collected telemetry feeds four analysis modules whose outputs are de-duplicated and ranked by expected monthly savings.

4.3 Covering surges: supplementary reservation

Surges occur only a fraction ρ of the time (Equation (1)). A unit of *surge* capacity therefore has utilization $u = \rho$. Applying Proposition 2 with the supplementary-contract discount δ_s yields the surge-coverage policy.

Proposition 3 (Supplementary reservation for surges). *It is optimal to cover a unit of surge capacity with a supplementary reservation (rather than on-demand) if and only if the surge time-fraction satisfies*

$$\rho \geq 1 - \delta_s. \quad (5)$$

Consequently, longer and/or more frequent surges (larger ρ) make supplementary reservation more attractive, and deeper supplementary discounts (larger δ_s) lower the surge-frequency threshold at which reservation pays off.

This recovers, in a transparent first-order form, the comparative statics of the richer dynamic model of Chen et al. (2024): supplementary reservation becomes more valuable as average surge and inter-surge durations grow and as the discount rate increases. The full dynamic problem additionally trades off cancellation fees and renewal timing, leading to their two-threshold policy; our static condition (5) is the break-even that such a policy oscillates around.

4.4 Spot capacity: an interruptible third tier

Spot instances are cheaper still ($s < r$) but interruptible. Serving a unit of load “spot-first” incurs cost s with probability $1 - \beta$ and falls back to on-demand at cost p with probability β .

Proposition 4 (Spot dominance and the reliability cap). *The expected unit cost of spot-first service is $\bar{c}_{\text{spot}} = (1 - \beta)s + \beta p$, and $\bar{c}_{\text{spot}} < p$ for every $s < p$ and $\beta < 1$. Hence spot weakly dominates on-demand in expectation for any interruptible load. If a fraction α of demand requires uninterrupted service (an availability SLA), then spot may serve at most the interruptible fraction $1 - \alpha$, and the cost-minimizing layering is: reserved base Q^* (Proposition 1), spot for interruptible residual demand, and on-demand for the reliability-constrained remainder.*

Proposition 4 formalizes why production systems should aggressively shift *interruptible* surge load to spot while protecting an SLA-critical core with reserved and on-demand capacity.

5 The Smart Cloud Optimizer System

The policies of Section 4 require two inputs that a real deployment must produce: the demand distribution F (and its forecast), and a mapping from abstract “capacity units” to concrete instance types. *Smart Cloud Optimizer* supplies both. Figure 1 sketches the architecture.

5.1 Demand forecasting

The forecasting module fits five models to each daily cost/demand series and selects the best by cross-validated error: (i) *Naive* (last value); (ii) *Seasonal naive* with weekly period $m = 7$; (iii) *ETS* (Holt–Winters) with additive trend and additive weekly seasonality ($m = 7$); (iv) *Prophet* with weekly and yearly seasonality; and (v) *SARIMAX* with auto-selected orders over a weekly season. Model selection uses *walk-forward* (expanding-window) cross-validation with an initial training window of 60 days, a forecast horizon of 14 days, and a step of 7 days, scored by MAPE, RMSE, and MAE. The selected model produces both point forecasts and predictive intervals; the latter yield the demand distribution F consumed by Proposition 1. An anomaly-detection routine (rolling z -score with window 30 and threshold 3σ , combined with a $1.5\times$ IQR rule) flags surge days for separate treatment under Proposition 3.

5.2 Right-sizing as a mixed-integer program

Given per-instance utilization telemetry, the right-sizing module selects, for each running instance i in the fleet $\mathcal{I}$, a target instance type j from a catalog $\mathcal{J}$ so as to minimize cost subject to meeting a utilization-derived capacity requirement. Let $x_{ij} \in \{0, 1\}$ indicate that instance i is mapped to type j , let c_j be the (on-demand) monthly price of type j , and let v_j, m_j be its vCPU and memory capacities. The program is

$$\min_x \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} c_j x_{ij} \quad (6)$$

$$\text{s.t.} \quad \sum_{j \in \mathcal{J}} x_{ij} = 1, \quad \forall i \in \mathcal{I}, \quad (7)$$

$$\sum_{j \in \mathcal{J}} v_j x_{ij} \geq V_i, \quad \forall i \in \mathcal{I}, \quad (8)$$

$$\sum_{j \in \mathcal{J}} m_j x_{ij} \geq M_i, \quad \forall i \in \mathcal{I} \text{ (when defined)}, \quad (9)$$

$$\sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} c_j x_{ij} \leq B, \quad \text{(optional budget cap)}, \quad (10)$$

$$x_{ij} \in \{0, 1\}.$$

The capacity requirement is derived from observed load with a safety margin: $V_i = \frac{\pi_i}{100} v_{c(i)} h$, where π_i is the 95th percentile of instance i ’s CPU utilization, $v_{c(i)}$ is its current vCPU count, and $h = 1.3$ is a 30% headroom factor; M_i is defined analogously when a memory signal is available. Constraint (7) assigns exactly one type per instance, (8)–(9) enforce sufficient capacity, and (10) optionally caps total spend. The program is solved with the CBC branch-and-cut solver. Right-sizing (choosing j) and the pricing-plan decision (reserved vs. on-demand, governed by Proposition 2) are complementary: the MILP fixes the resource, and the threshold policy fixes its purchasing mode.

5.3 Rule-based checks and an LLM advisor

Beyond compute right-sizing, ten heuristic checks detect service-specific waste, e.g. switching long-running on-demand instances to reserved (the empirical analogue of Proposition 2), down-sizing over-provisioned serverless memory, upgrading storage volume classes, deleting unattached or idle resources, switching database capacity modes, and replacing costly network gateways with endpoints. Each check emits a recommendation with an estimated monthly saving and a confidence

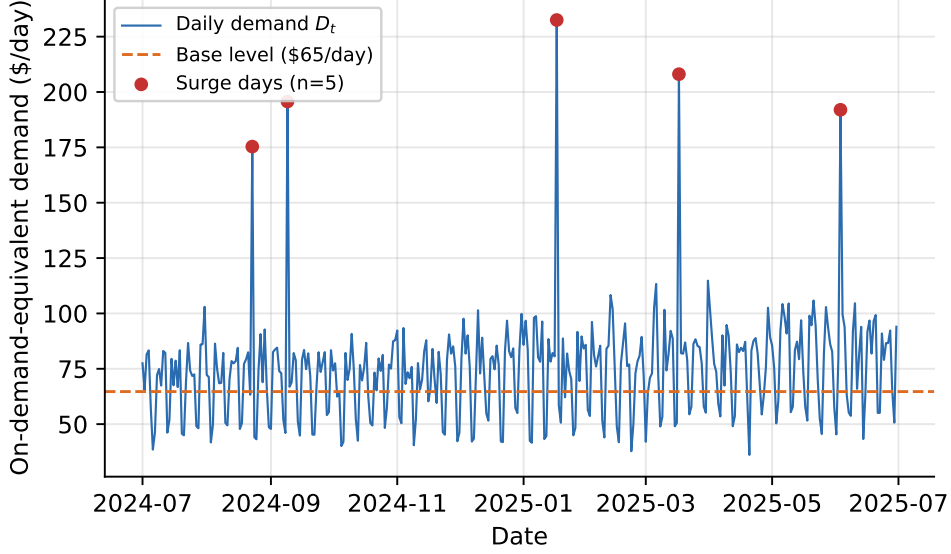


Figure 2: Daily demand over one year: a seasonal base (dashed) punctuated by five intermittent surges (red).

label, and duplicates across modules are resolved by keeping the highest-saving recommendation per resource. Finally, for *greenfield* provisioning (no telemetry yet), a large-language-model advisor maps a short natural-language requirements profile to a concrete, cost-optimized architecture with a cost estimate.

6 Numerical Study

We evaluate the theory and the system on a year-long workload drawn from the system’s reference dataset (account `aws-SYNTHETIC-001`), comprising daily costs and hourly utilization across eight EC2 instances and nine auxiliary services. The dataset is synthetic but constructed to be realistic; all numbers below are regenerated by `paper/numerical_study.py`.

6.1 Demand exhibits the base-plus-surge structure

Figure 2 plots the daily demand series. Over $T = 365$ days (2024-07-01 to 2025-06-30) demand averages \$74.6/day (median \$76.0, std \$22.8, coefficient of variation 0.31) on a base level of \$64.7/day. It displays pronounced *weekly seasonality*—weekday mean \$84.4 versus weekend mean \$50.1—and *intermittent surges*: five days (1.4% of the horizon) exceed the surge threshold of \$121.7 (median $+2\sigma$), peaking at \$232.5, i.e. $3.06\times$ the median. This is precisely the structure of Equation (1), and it validates the modeling premise of Chen et al. (2024) on concrete data.

6.2 Forecasting accuracy

Table 2 reports walk-forward cross-validation over 42 folds. Seasonal models dominate the naive baseline by more than a factor of two in MAPE, confirming that the weekly structure is both real and learnable. Exponential smoothing (ETS) is the most accurate (MAPE 12.1%), narrowly ahead of SARIMAX, and is selected as the production model. Figure 3 shows a representative forecast.

Table 2: Forecasting accuracy (walk-forward CV: initial 60d, horizon 14d, step 7d; 42 folds). Lower is better; best in **bold**.

Model	MAPE (%)	RMSE (\$)	MAE (\$)
Naive	30.04	24.73	18.65
Seasonal naive	15.57	19.50	12.46
SARIMAX(1, 1, 1)(1, 0, 1) ₇	12.50	14.95	9.81
ETS (Holt–Winters, $m=7$)	12.12	14.66	9.64

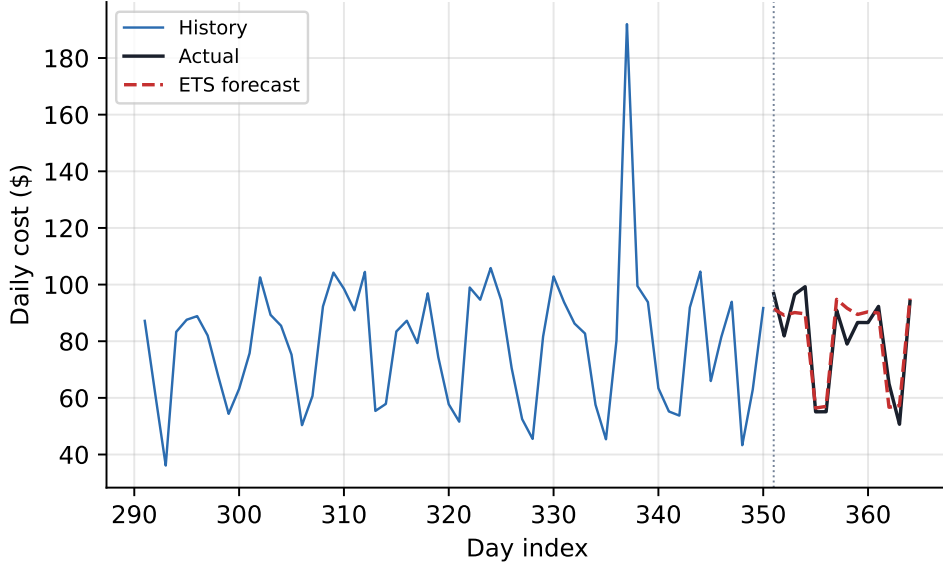


Figure 3: ETS forecast versus actual on the final 14-day horizon (history in blue, actual in black, forecast in red).

6.3 The newsvendor policy: optimum and savings

We now apply Proposition 1 to the empirical demand distribution. Figure 4 shows the empirical CDF with the optimal reservation levels marked, and Figure 5 shows the (convex) annual-cost curve $C(Q)$ with its minimizer. Table 3 compares procurement strategies.

Three findings stand out. First, the closed-form reservation level (3) coincides with exhaustive grid search to within \$0.03/day, empirically confirming Proposition 1; at $\delta = 0.37$ the optimum reserves the 37th demand percentile, and the realized coverage fraction is $F(Q^*) = 0.370$, exactly as Corollary 1 predicts. Second, the optimal policy reduces annual cost by 26.7% (one-year) to 50.0% (three-year) relative to pure on-demand, while a naive “reserve to cover the peak” policy is 60–63% *more* expensive than the optimum because it pays for capacity that is idle on all non-surge days. Third, as the discount deepens the optimum reserves a higher quantile (\$70.4 $\rightarrow$ \$80.7), illustrating Corollary 1.

6.4 Sensitivity to the discount rate

Figure 6 sweeps the discount δ from 0.05 to 0.80. Both the optimal reserved level $Q^* = F^{-1}(\delta)$ and the resulting savings rise monotonically with δ , consistent with Corollary 1: the value of commitment, and the amount one should commit, grow with the discount. The break-even utilizations

Table 3: Annual cost (\$) and savings under different reservation strategies, relative to provisioning all demand on-demand (\$27,243). Q^* closed form vs. exhaustive grid search.

Tier	δ	Q^* (\$/day)	grid Q^*	Optimal cost	Save vs. on-demand	Save vs. reserve-peak
On-demand only	—	—	—	27,243	—	—
RI 1-year	0.37	70.36	70.38	19,975	26.7%	62.6%
RI 3-year	0.60	80.70	80.70	13,624	50.0%	59.9%

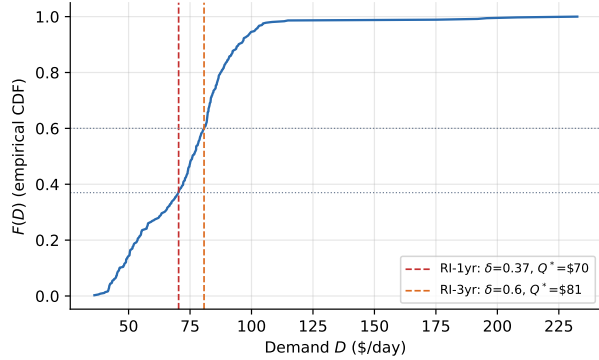


Figure 4: Empirical demand CDF; optimal reservation levels at the δ -quantiles.

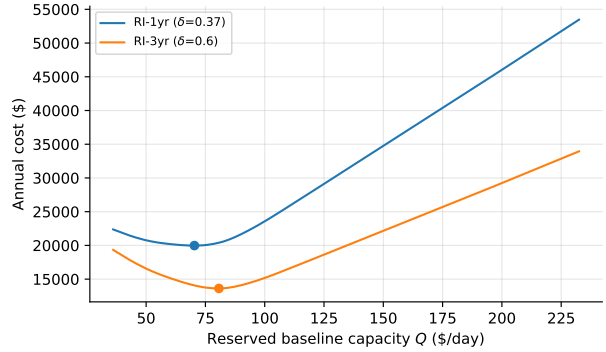


Figure 5: Convex annual-cost curve $C(Q)$; markers at the optimum Q^* .

of Proposition 2 ($u^* = 0.63$ and 0.40 at the two real tiers) are marked for reference.

6.5 End-to-end case study

Finally we run the full system on the same account. The optimizer produces 19 recommendations spanning seven action types plus 10 greenfield LLM recommendations, identifying \$590.4/month in savings against an addressable spend of \$1,074.8/month—a 54.9% reduction on the optimized scope (Table 4). Compute actions dominate: right-sizing (\$362.8/month) and pricing-plan switches (\$208.4/month, the empirical analogue of Proposition 2) together account for 97% of the savings. The fleet’s implied discount tiers recovered from the pricing table—37.0% (1-yr), 60.0% (3-yr), 70.0% (spot)—are exactly the δ values used in our analysis, so the model is parameterized by the deployment’s own prices.

7 Managerial Insights

Our analysis and experiments yield several actionable insights.

1. **Reserve to a demand quantile, not to the peak or the mean.** The cost-minimizing reserved capacity is the δ -quantile of demand (Proposition 1). Reserving for the peak wastes 60–63% of the achievable optimum in our data; reserving the mean is close but systematically sub-optimal.
2. **Reserve more when discounts are deeper.** Both the optimal reservation level and the realized savings increase with the discount δ (Corollary 1, Figure 6). Three-year commitments are warranted for the stable base; shorter commitments suit volatile components.

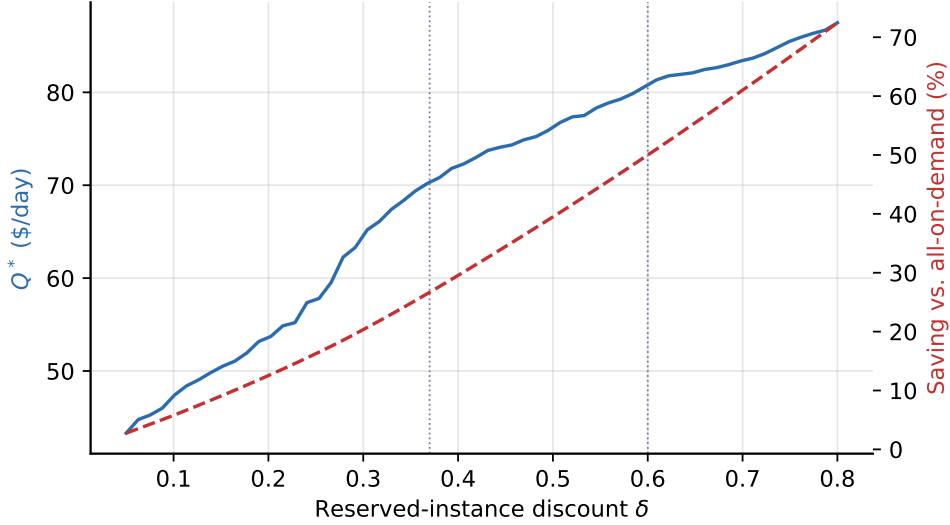


Figure 6: Optimal reserved level (left axis) and cost saving versus on-demand (right axis) as functions of the reserved discount δ . Dotted lines mark the real 1-year and 3-year tiers.

Table 4: System recommendations on **aws-SYNTHETIC-001**: count and monthly savings by action type.

Action type	#	Monthly savings (\$)
Right-sizing (MILP)	2	362.81
Pricing-plan switch (reserve)	6	208.42
Delete unused resource	3	10.00
Replace with VPC endpoint	2	5.67
Storage volume-type upgrade	4	2.60
Storage-class switch	1	0.79
Serverless memory resize	1	0.11
Total	19	590.40

3. **Replace the “60-day” heuristic with a utilization threshold.** A workload should be reserved exactly when its utilization exceeds $u^* = 1 - \delta$ (Proposition 2)—63% for one-year and 40% for three-year RIs—rather than when it has merely been running for a fixed number of days.
4. **Match surge coverage to surge frequency.** Supplementary reservation of surge capacity pays off only when the surge time-fraction exceeds $1 - \delta_s$ (Proposition 3); rare, spiky surges should ride on-demand or interruptible spot capacity, not standing reservations.
5. **Forecast the distribution, then optimize.** Accurate seasonal forecasting (ETS at 12.1% MAPE) is what makes the quantile in (3) trustworthy; the forecasting and reservation steps are complementary, not substitutes.

8 Conclusion

We connected the operations-management theory of capacity reservation to the engineering practice of cloud cost optimization. Modeling a workload as a seasonal base plus intermittent random surges, we derived a newsvendor reservation level, a utilization-threshold purchasing rule that subsumes a widespread industry heuristic, and surge- and spot-coverage conditions. We embedded these policies in *Smart Cloud Optimizer*, an end-to-end system that forecasts demand, right-sizes via mixed-integer programming, and ranks savings. On a realistic year-long workload the theory matched exhaustive search, forecasting achieved 12.1% MAPE, and the system surfaced \$590/month (54.9%) in savings.

Several limitations point to future work. Our evaluation uses a single synthetic-but-realistic account; validating on production multi-account telemetry is the natural next step. The reservation analysis is static given a forecast distribution; integrating the dynamic purchase/renew/cancel dynamics—including cancellation fees and a secondary reservation marketplace, as in [Chen et al. \(2024\)](#)—into the running system would close the loop between theory and deployment. Finally, jointly optimizing right-sizing and the pricing-plan mix in a single program, rather than sequentially, is a promising extension.

Reproducibility. All figures and numbers in Section 6 are regenerated by `paper/numerical_study.py` against the project database.

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A Proofs

Proof of Proposition 1. Write $g(Q) = \mathbb{E}[c(Q)] = (1 - \delta)Q + p \mathbb{E}(D - Q)^+$ with $p = 1$. The map $Q \mapsto (D - Q)^+ = \max\{D - Q, 0\}$ is convex for each realization of D , and expectation preserves convexity; adding the linear term $(1 - \delta)Q$ keeps g convex. For the derivative, since $\frac{\partial}{\partial Q} (D - Q)^+ = -\mathbf{1}\{D > Q\}$ almost everywhere,

$$g'(Q) = (1 - \delta) - \Pr(D > Q) = (1 - \delta) - (1 - F(Q)) = F(Q) - \delta.$$

g' is nondecreasing (as F is), confirming convexity, and the first-order condition $g'(Q^*) = 0$ gives $F(Q^*) = \delta$, i.e. $Q^* = F^{-1}(\delta)$. The newsvendor reading follows by identifying the underage cost $C_u = p - r = \delta$ (extra cost per unit served on-demand instead of reserved) and the overage cost $C_o = r = 1 - \delta$ (cost per idle reserved unit), whence the critical ratio $C_u/(C_u + C_o) = \delta$. $\square$

Proof of Corollary 1. F^{-1} is nondecreasing, so $Q^*(\delta) = F^{-1}(\delta)$ is nondecreasing in δ . The coverage identity $F(Q^*) = \delta$ is the first-order condition from Proposition 1. $\square$

Proof of Proposition 2. Over one reservation term, reserving a unit costs $r = (1 - \delta)p$ with certainty. Serving the same unit on-demand costs p for each period it is active; with utilization u the expected term cost is pu . Reservation is preferred iff $(1 - \delta)p \leq pu$, i.e. $u \geq 1 - \delta$. Substituting $\delta \in \{0.37, 0.60\}$ gives $u^* \in \{0.63, 0.40\}$. $\square$

Proof of Proposition 3. A unit of surge capacity is active exactly during surge periods, hence its utilization is $u = \rho$. Applying Proposition 2 with the supplementary discount δ_s gives the reservation condition $\rho \geq 1 - \delta_s$. Monotonicity in ρ and δ_s is immediate. $\square$

Proof of Proposition 4. With probability $1 - \beta$ the spot request is served at cost s ; with probability β it is reclaimed and re-served on-demand at cost p , so the expected unit cost is $\bar{c}_{\text{spot}} = (1 - \beta)s + \beta p$. Then $p - \bar{c}_{\text{spot}} = (1 - \beta)(p - s) > 0$ whenever $s < p$ and $\beta < 1$, establishing dominance. Under an availability SLA requiring a fraction α of demand to be uninterrupted, only the interruptible fraction $1 - \alpha$ is eligible for spot; minimizing cost subject to this eligibility constraint and to reserved capacity being sunk yields the stated layering (reserved $\rightarrow$ spot $\rightarrow$ on-demand). $\square$